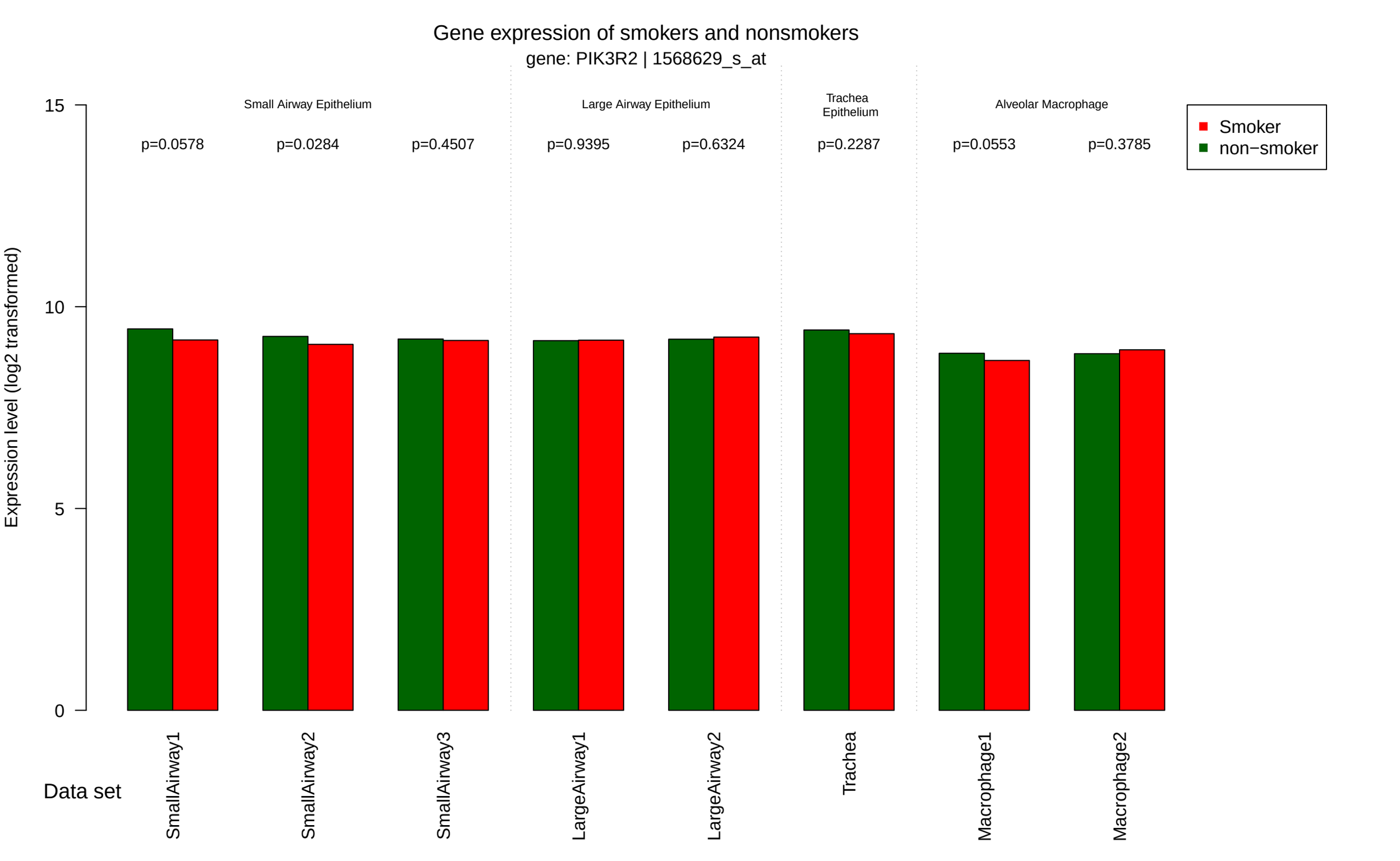
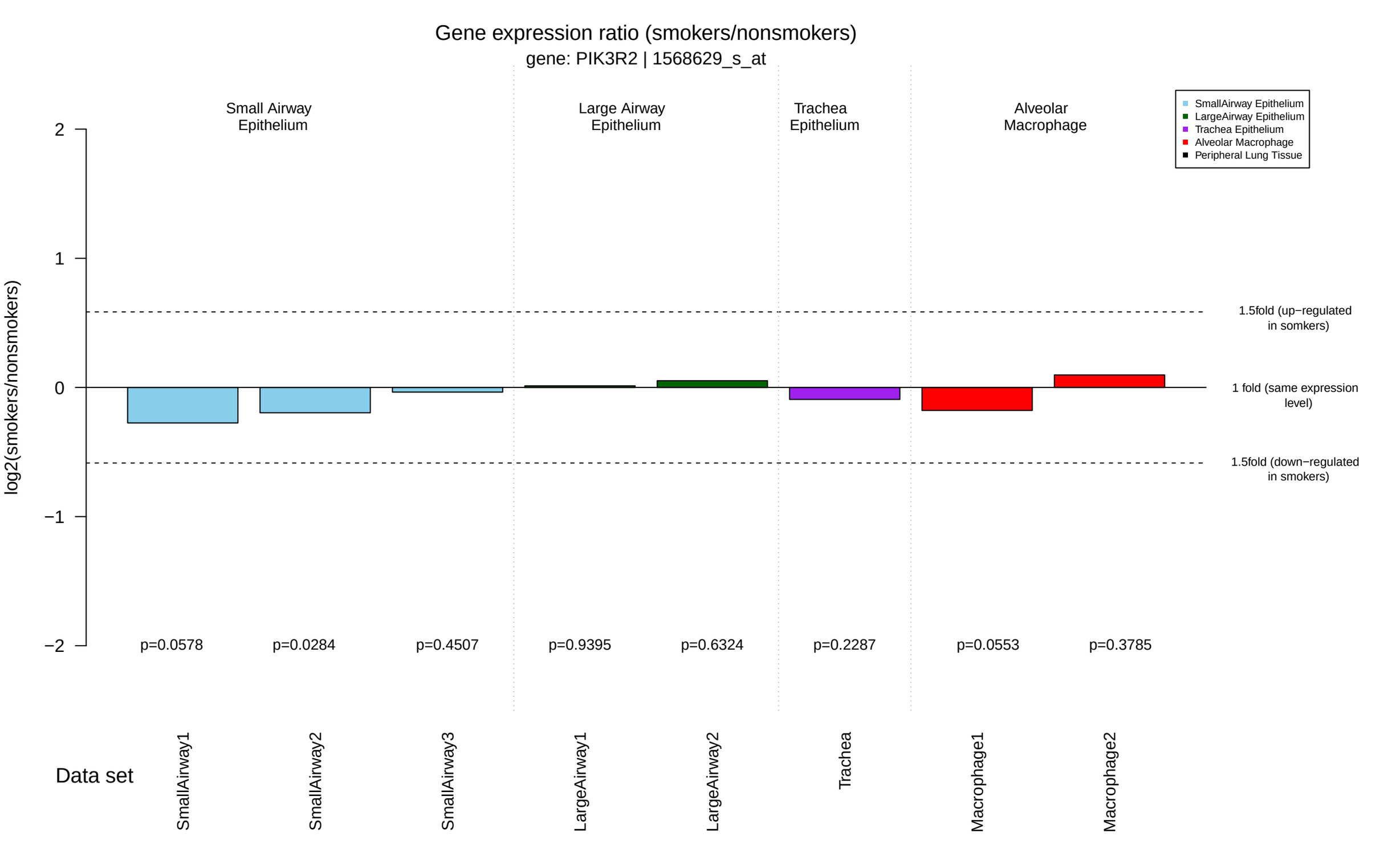
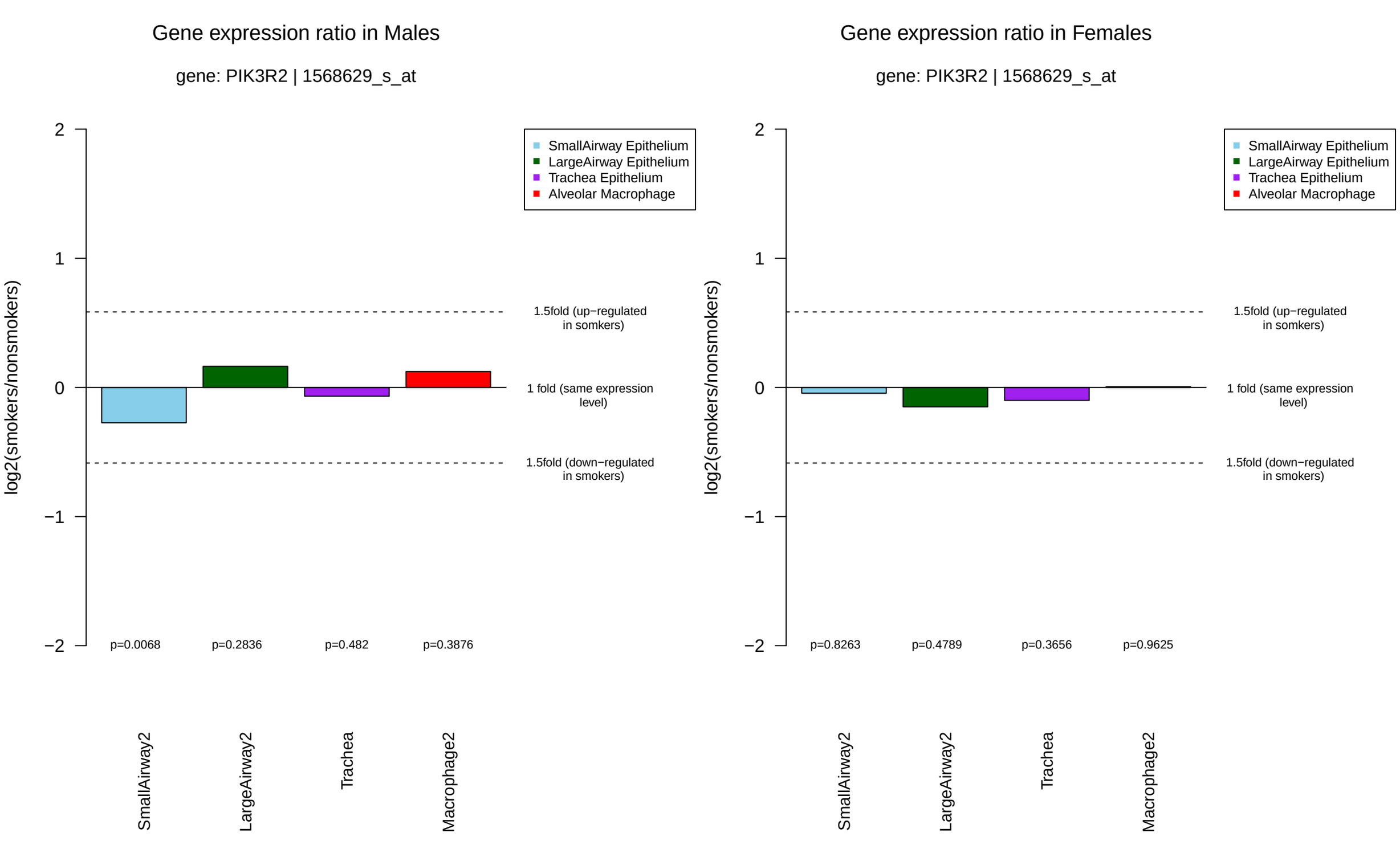
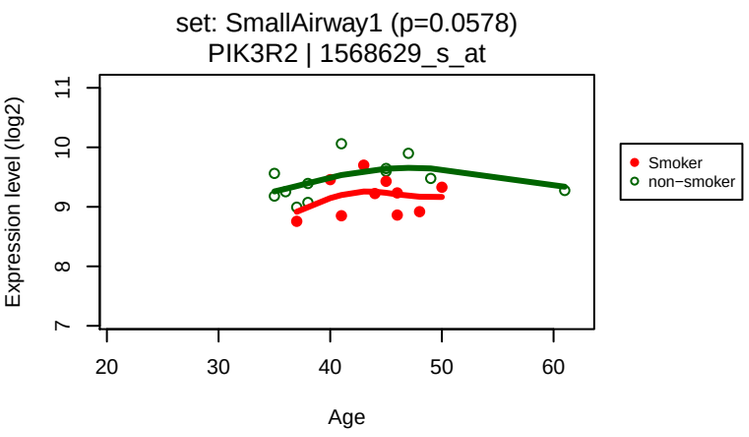
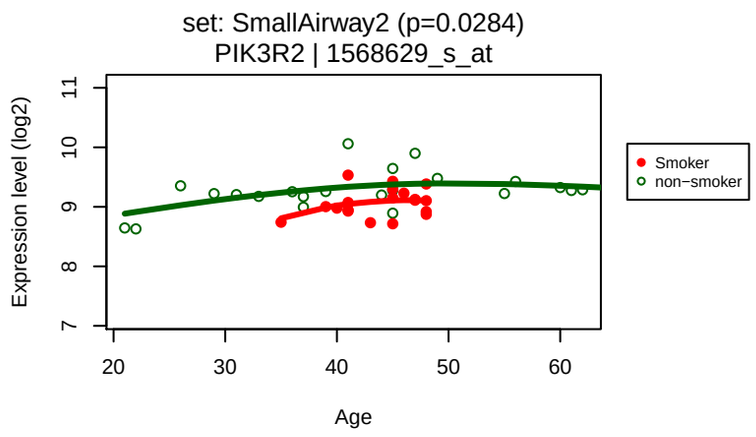
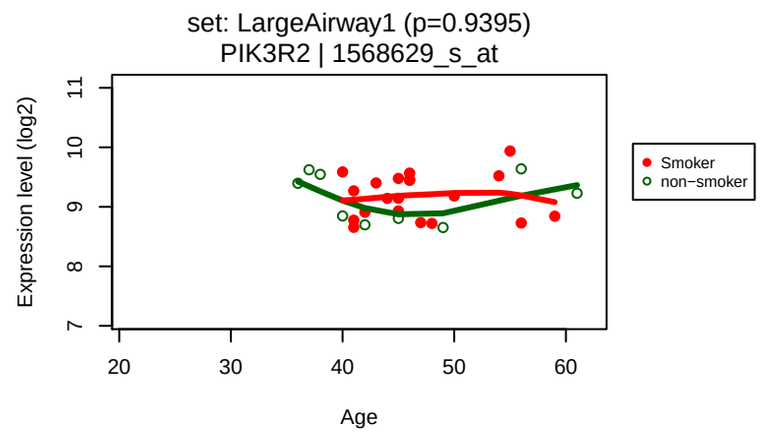

Gene expression of smokers and nonsmokers
gene: PIK3R2 | 1568629_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0578
p=0.0284
p=0.4507
p=0.9395
p=0.6324
p=0.2287
p=0.0553
p=0.3785
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

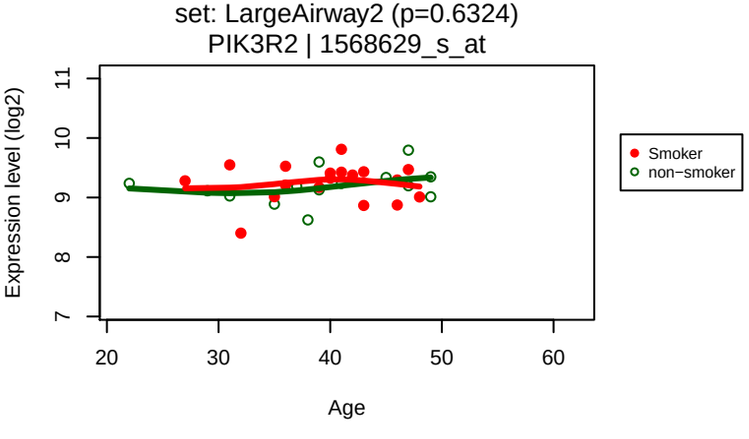

Gene expression ratio (smokers/nonsmokers)
gene: PIK3R2 | 1568629_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0578
p=0.0284
p=0.4507
p=0.9395
p=0.6324
p=0.2287
p=0.0553
p=0.3785
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

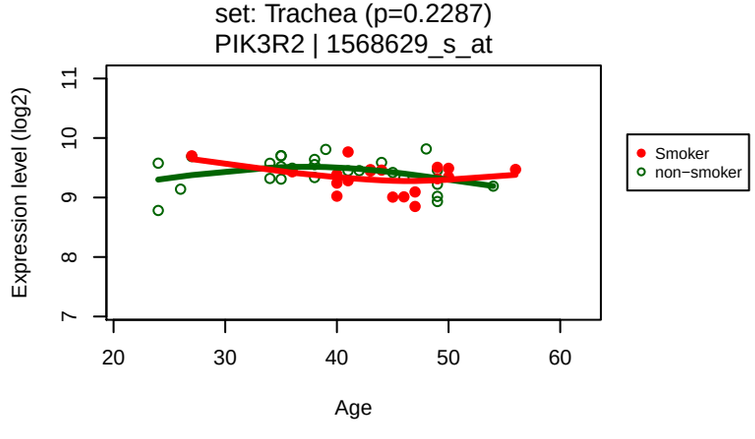

Gene expression ratio in Males
gene: PIK3R2 | 1568629_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0068
p=0.2836
p=0.482
p=0.3876
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3R2 | 1568629_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8263
p=0.4789
p=0.3656
p=0.9625
SmallAirway2
LargeAirway2
Trachea
Macrophage2

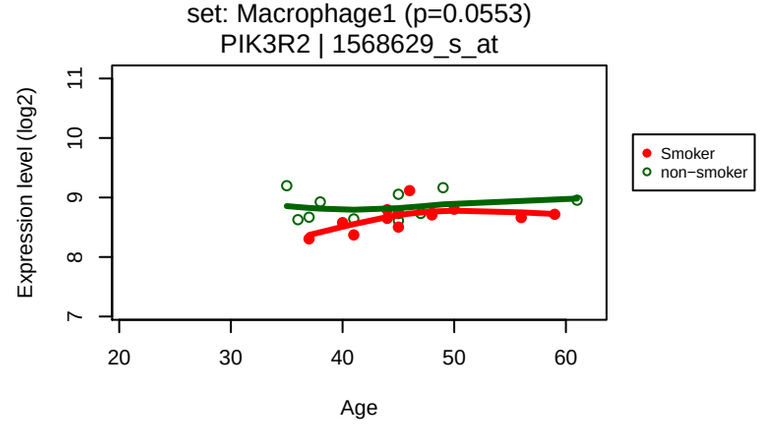

set: SmallAirway1 (p=0.0578)
PIK3R2 | 1568629_s_at
Expression level (log2)
Age
Smoker
non−smoker

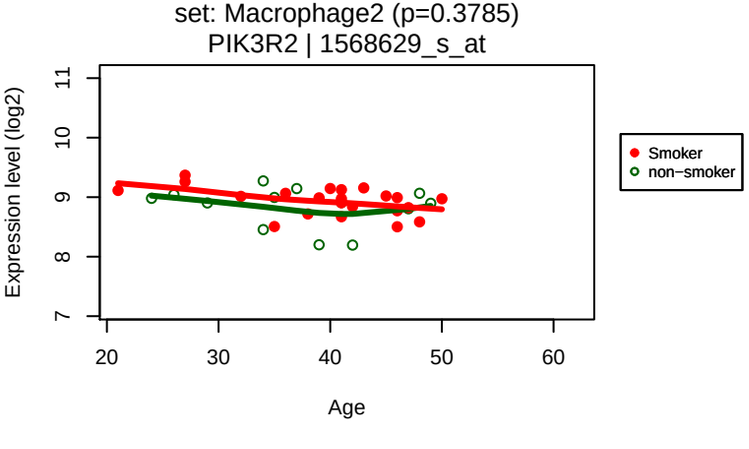

set: SmallAirway2 (p=0.0284)
PIK3R2 | 1568629_s_at
Expression level (log2)
Age
Smoker
non−smoker

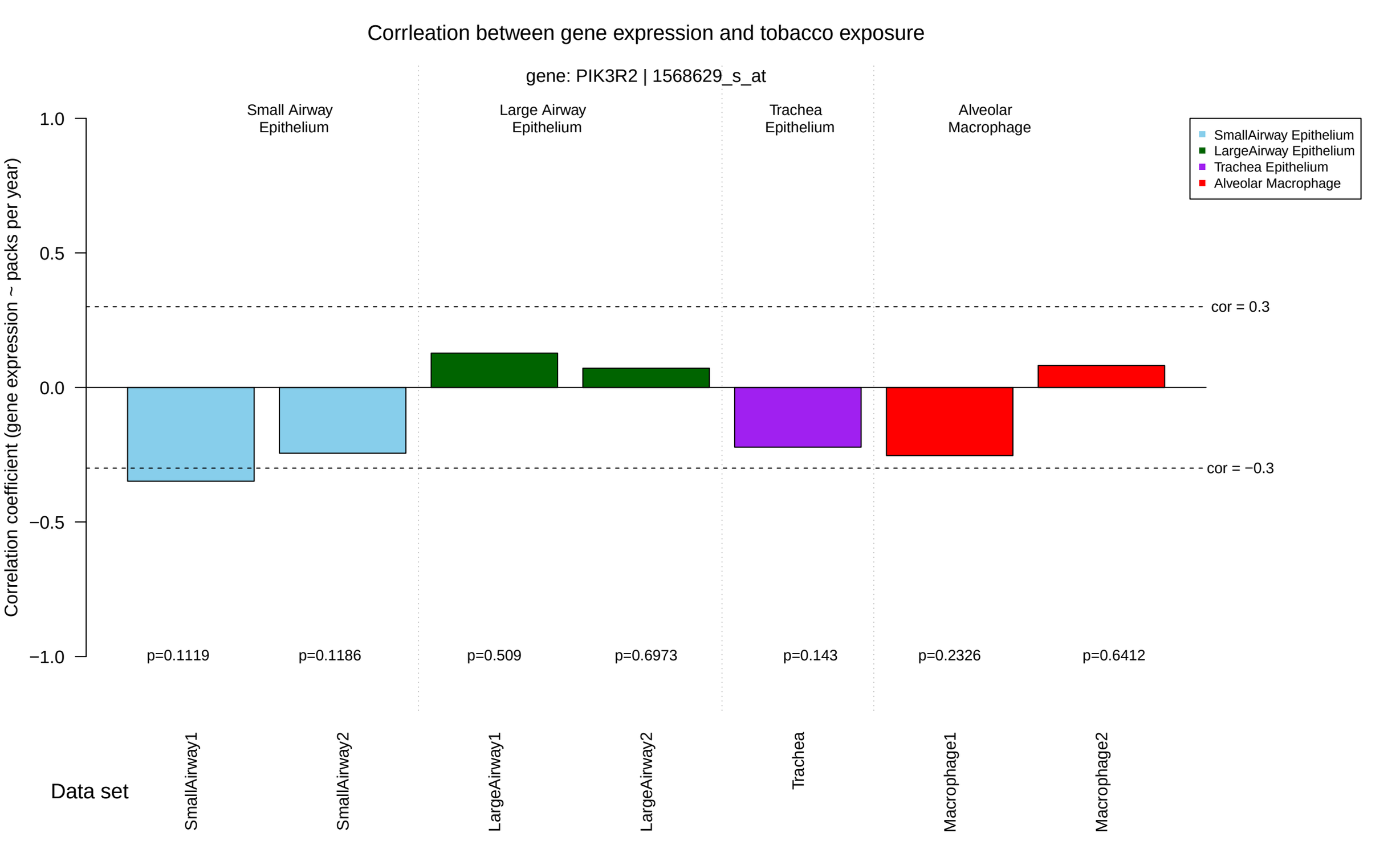

set: LargeAirway1 (p=0.9395)
PIK3R2 | 1568629_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6324)
PIK3R2 | 1568629_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2287)
PIK3R2 | 1568629_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0553)
PIK3R2 | 1568629_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3785)
PIK3R2 | 1568629_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3R2 | 1568629_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1119
p=0.1186
p=0.509
p=0.6973
p=0.143
p=0.2326
p=0.6412
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2